

Market Scenario of Some Key Consumer Electronics Products

Products	Details
Colour TVs (CTVs)	- CTVs are the largest contributors to this segment - TV industry in India is estimated to have reached ₹ 660 billion (US\$ 10.19 billion) in CY17 and projected to reach ₹ 862 billion (US\$ 13.31 billion) in CY20 - Number of TV households in India is expected to reach 200 million by 2020
Flat panel display	- LED/LCD/plasma TV sets present a huge opportunity in India with a penetration of only fourteen per cent households - Production of LCD/LED TVs has also grown at a fast pace in India from 8.7 million units in FY15 to 16 million units in FY18 - Price decline due to relatively low import duty on LCD panels, higher penetration levels and introduction of small entry-size models are key growth drivers in the segment
Direct-to-home (DTH)	- The set-top box (STB) market is growing rapidly due to expansion of DTH and introduction of conditional access system (CAS) in metros - Active DTH subscribers in the country increased to 69.45 million in September 2018 from 63.61 million in March 2017
Refrigerators	- This segment made up 27 per cent of the consumer appliances market in 2017 - Market share of direct cool and frost-free segment in FY18 is estimated at seventy per cent and thirty per cent, respectively. - Refrigerator market in India was estimated at around ₹ 195 billion (US\$ 3.03 billion) and is expected to reach ₹ 344 billion (US\$ 5.34 billion) by 2022. Production of refrigerators in India increased three per cent year-on-year in FY18 (up to February 2018) to 11.88 million units
Air-conditioners (ACs)	- Installed stock of room ACs in India increased from two million units in 2006 to thirty million units in 2017 and is expected to be between 55 and 124 million by 2030. - As of April 2019, AC makers expect a double-digit rise in sales and adoption of superior technology and power-efficient range of inverter ACs by consumers - During FY18 (up to February 2018) production of ACs in India increased 4.9 per cent year-on-year to 3.19 million units. Split ACs had a 85 per cent market share in FY18. Overall room ACs market in the country is expected to increase to seven million units by FY21 from 5.5 million units in FY18. - Daikin, air conditioner maker expects a twenty per cent growth in sales and aims to become a ₹ 50 billion (US\$ 715.41 million) company in FY20 backed by rising demand of power-efficient inverter ACs and sales network expansion
Washing appliances	Production of washing appliances in India increased 5.1 per cent year-on-year in FY18 (up to February 2018) to 5.79 million units. Washing machine market in India was estimated at around ₹ 70 billion (US\$ 1.09 billion) in FY18
Electric fans	- Domestic electric fan market in India increased at thirteen per cent CAGR between FY08-17 to reach ₹ 70 billion (US\$ 1.1 billion). Around 22.6 million fans were produced in the country in FY18 (up to February 2018) - Penetration of electric fans in rural areas is expected to reach 76 to 78 per cent in 2019-20 from 65 per cent in 2017

(Source: CEAMA, Electronic Industries Association of India, Mirc annual report, Edelweiss Investment Research, CMIE, Intex Technologies, TERI- Improving Air Conditioners in India, Motilal Oswal, IBEF)

ing innovative electronic devices, but technical challenges of joining conductive materials to 3D-printed circuit materials must be addressed to explore the full potential of this technology in the consumer electronics sector.

A more advanced technology, 4D printing involves using smart or shape-memory materials to print objects that can self-assemble into multiple configurations. This tech-

nology is still in the nascent stage but poses enormous potential to revolutionise the consumer electronics sector.

Moving forward

The consumer electronics market is always flooded with latest technologies and innovations. However, product lifecycle is quite short, as consumer interests change fast. Thus, many technologies and innovations

often end up disappearing within a short span of time. The dividing line between maturing the technology for widespread use and its obsolescence due to new technology introduction is razor-thin. Thus, speed in manufacturing and launching the products is critical in this sector.

In addition, delivering innovative products that are smaller, faster, more durable, more affordable and have a better battery life, as per ever-increasing consumer demands, can only be achieved through successful adoption of innovative technologies and use of innovative materials. Thus, a continuous focus on R&D is also must to achieve a competitive edge.

Considering the importance of environmental issues, the consumer electronics segment needs to focus more and more on electronic waste (e-waste) management. India produces two million tonnes of e-waste every year, but no proper guidelines are available from the government's side. Majority of brands operating in India do not have a tangible extended producer responsibility (EPR) in place for taking back or managing their end-of-life waste electronics and electrical equipment (WEEE).

In some cases, though a take-back system is present but that is not functional. Most OEMs do not provide adequate information on their website about any take-back or recycling programme.

Moreover, India being a vast country, setting up collection mechanisms is a big challenge.

Another major issue in e-waste management in India is how to include thousands of producers and importers under the ambit of regulation.

However, industry cooperation with correct intervention from the government can help the industry manage the e-waste menace. India being highly-deficient in precious mineral resources, there is a need for a well-designed, robust and regulated e-waste recovery regime, which would generate employment, revenue as well as support sustainable development of the consumer electronics sector in the country. **EFY**